Bass Model analysis:

Samsung Micro RGB TV (2025) vs. LG OLED TV (2013–2025)

I chose the Samsung micro RGB TV from the TIME Best Inventions of 2025 list: https://time.com/collections/best-inventions-2025/7318277/samsung-micro-rgb/, knowing for sure that I'll find a look-alike product with similar innovation with many common traits/info about it.

- Since most of the companies do not openly share their sales/demand related data (time series data), I decided to scrape data from official articles from reliable publishers to put-together a realistic estimate of the look-alike product which in this case is LG OLED TV.

Part 1: 2025 Innovation: Samsung Micro RGB TV

Samsung launched the world's first Micro RGB TV in August 2025 as a 115-inch model priced at $29.999, with expanded sizes (55" to 130") announced at CES 2026.

Unlike normal premium TVs that rely on a single white or blue LED backlight filtered through color layers, Samsung's Micro RGB technology uses individually controlled red, green, and blue LEDs each less than 100 micrometers wide, arranged in an ultra-fine pattern behind the LCD panel. This architecture enables precision per-LED color control, delivering richer and more lifelike colors, near-perfect black levels through full-color local dimming, and a 20% reduction in power consumption. An on-board AI engine further optimizes each frame in real time. Samsung's proprietary Glare Free glass coating completes the package.

The Micro RGB TV currently targets the *ultra-premium* customer segment, competing directly with **OLED and MicroLED** displays. It is certified at 100% BT.2020 wide color gamut by VDE, and Samsung positions it as its flagship display innovation for 2025–2026.

Part 2: Look-Alike Product: LG OLED TV (2013–2025)

Product description

The LG OLED TV, first launched commercially in 2013 by LG Electronics, is selected as the look-alike product. LG pioneered consumer OLED television using LG Display's WOLED (White OLED) panel technology with color filters, establishing a new premium TV category that did not exist before.

- By the way, LG has been the world's #1 OLED TV brand for over 11 consecutive years (Omdia, 2013–2024).

Justification for Look-Alike Pairing

Samsung Micro RGB TV and LG OLED TV share deep structural parallels that make this a strong product-to-product pairing for Bass model estimation:

- Both are premium TV display innovations that introduced fundamentally new backlight/emitter architectures to the consumer TV market.
  - LG OLED TV replaced LCD backlights with self-emitting organic diodes; Samsung Micro RGB replaces conventional LED backlights with micro-scale individually controlled RGB LEDs.
- Both launched at ultra-premium price points.

    - LG's first 55" OLED TV debuted at approximately $15,000 in 2013. Samsung's first 115" Micro RGB TV launched at $29,999 in 2025. Both relied on price declines, expanded size ranges, and manufacturing scale-up to drive adoption over time.
- Both targeted picture quality supremacy.
    - LG OLED TV marketed "perfect blacks" and infinite contrast from self-emitting pixels. Samsung Micro RGB markets 100% BT.2020 color gamut, per-LED RGB dimming, and near-indistinguishable-from-real-life imager.
- Both are TV-vs-TV category creators from major Korean manufacturers.
    - LG (with LG Display) created the consumer OLED TV category. Samsung is creating the consumer Micro RGB LED backlit TV category. Both compete for the premium large-screen TV buyer.
- Both show similar early adoption dynamics.
    - LG OLED TV experienced slow initial adoption (2013–2015) limited by high cost and small product range, followed by accelerating growth as prices fell and brand partners (Sony, Panasonic, Philips) joined. Samsung Micro RGB TV is expected to follow a similar pattern: ultra-premium early adopters in 2025, then broader adoption as sizes expand and prices decrease.

Bass Model Applicability

The LG OLED TV adoption curve from 2013 to 2025 provides 13 years of data showing a clear innovation-to-growth trajectory.

- The early years (2013–2016) reflect innovation-driven adoption by enthusiasts willing to pay a premium (innovation coefficient p).
- The rapid growth phase (2017–2021) reflects imitation-driven adoption as prices fell, product range expanded, and word-of-mouth from reviewers and early adopters drove mainstream interest (imitation coefficient q).
- The 2022–2023 dip and 2024–2025 recovery reflect market maturation and competitive pressure from MiniLED.

This data makes estimation of p, q, and m parameters applicable to forecasting Samsung Micro RGB TV's diffusion.

Part 3: Historical Data for the Look-Alike Product

Variable and Source

Variable: Global OLED TV Annual Unit Shipments (millions of units)
Time Period: 2013–2025 (13 years)
Geography: Worldwide

Once again:

- The data below is compiled from freely accessible industry press releases and published analyst reports. Individual year sources are detailed in the footnotes. All numbers represent actual shipment estimates (not forecasts) published by recognized display-industry research firms.
- No paid subscription (Statista, Omdia) was required; all info was gathered from free press releases, news articles, and published LG corporate communications.

Data Table

Year	Annual Shipments (Million Units)	Cumulative Shipments (Million Units)	Key Source	Confidence
2013	0.005	0.005	DisplaySearch: "in 2013 only about 4,500 OLED TVs were sold" (ref)	Direct
2014	0.077	0.082	DisplaySearch: "77,000 OLED TVs were sold worldwide in 2014" (ref)	Direct
2015	0.40	0.482	LGD conference call: "LGD shipped around 400,000 OLED TV panels in 2015" (ref)	Direct
2016	0.72	1.202	Verifiable at Statista (ref).	Direct
2017	1.47	2.672	IHS Markit: "exceed 1.4 million units shipped in 2017" (taken down ref | Verifiable at Statista (ref).	Direct
2018	2.51	5.182	IHS reported 19% OLED growth in 2019; 2019 actual = 3.0M → 2018 ≈ 3.0/1.19 = 2.52M. Cross-check: LGD shipped "2.2–2.8M panels" in 2018 (ref). ~2.51M is a reasonable midpoint	*Derived Number*
2019	3.00	8.182	IHS Markit: "Total OLED TV sales in 2019 reached 3 million units" (ref)	Direct
2020	3.94	12.122	TrendForce: 2021 = 6.7M, "70% growth YoY" → 2020 = 6.7/1.7 = 3.94M (ref). Cross-check: cumulative hit 10M in Sept 2020 (ref)	*Derived number*
2021	6.70	18.822	TrendForce: "OLED TV shipments to 6.7 million units in 2021" (TrendForce ref)	Direct
2022	6.67	25.492	TrendForce: "6.67 million units" (ref)	Direct
2023	5.44	30.932	TrendForce/Statista: "around 5.44 million units" (ref)	Direct
2024	6.07	37.002	Omdia via BusinessWire (ref)	Direct
2025	6.43	43.432	Omdia via OLED-Info (ref)	Direct

> The 2022–2023 deviation from the fitted curve is caused by macroeconomic factor of shocks (European inflation, energy costs) and also the competitive pressure from MiniLED, which are exogenous factors.

Part 4. Estimate Bass Model parameters.

> (completed in Rmd) To check scroll further down this report pdf).

- The NLS fit gave $p = 0.0032$, $q = 0.4428$ and $M \approx 59.718.200$ for the market potential.
- q/p ratio is ~138, meaning OLED TV adoption was almost entirely imitation-driven, which makes sense: one needed to see an OLED in person to get why it's worth the premium.

Part 5. Predict the diffusion of the innovation selected in step 1

> (completed, in Rmd) To check scroll further down this report pdf).

Part 6. Choose a scope (global or country-specific).

> (completed, in Rmd) To check scroll further down this report pdf).

Part 7. Fermi Estimation for Samsung Micro RGB TV Market Potential

The global TV market ships about 200–230M units/year (TrendForce 2023; Counterpoint 2025).

- Of those, roughly 8–10% fall in the premium segment above $1,500 — around 18M units/year.
- Our NLS fit estimates OLED TV's lifetime market potential at ~60M cumulative units.

Samsung Micro RGB TV targets the same premium segment, but must probably lead to steeper upward trend:
it launched at $29,999 (about 2x the first LG OLED's $15,000 in 2013), Samsung is the sole manufacturer (vs. 20+ OLED TV brands by 2022), and it enters a market where OLED and MiniLED are already mature competitors and the customers' perceived benefit is not going to be the same as for the OLEDs (my opinion).

So I scale OLED's m down by 0.65 to account for all of the above-mentioned; it i s a number which adds my personal opinion to the Fermi Estimate. On one hand this is a big penalty, on the other hand, one can argue that Samsung is an established brand for the ultra-premium segment and because of this, the coefficient may fluctuate as well: 0.55-0.75 (this also reflects my personal opinion).

Sources:

- TrendForce (Nov 2023) — ~197M TV shipments:
  - https://www.trendforce.com/presscenter/news/20231128-11931.html
- Counterpoint/Omdia (2025) — 230M units in 2024:
  - https://displaydaily.com/tv-market-recovering-getting-bigger-brighter-better/
- LG OLED launch price ($15K, 2013):
  - https://www.oled-info.com/lgs-oled-tv-will-arrive-uk-july-2013-9999-15000
- 20+ OLED TV brands by 2022:
  - https://www.lg.com/global/newsroom/lg-story/beyond-news/lg-oled-tv-a-decade-at-the-top-and-no-sign-of-slowing-down/

What I noticed during the Bass Model estimation

- **2013–2016 (p-driven):**
  - Tiny volumes (5K → 720K). Ultra-premium pricing ($5K–$15K), LG was basically the only player. The small p reflects this.
- **2017–2021 (q takes over):**

    - Sony, Panasonic, Philips joined using LG Display panels. Prices dropped below $2,000. Shipments jumped from 1.5M to 6.7M — classic imitation-driven growth.
- **2022–2025 (near peak, plateau):**
    - Shipments oscillated at 5.4–6.7M. Samsung entered with QD-OLED. The Bass model sees this as the neighborhood of peak adoption, which checks out.

With m ≈ 60M and 43.4M already shipped by 2025, OLED is ~73% through its total diffusion — well into late majority territory